HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Department of Actuarial Mathematics and Statistics

F70LB

F70LB LIFE INSURANCE MATHEMATICS B

Semester 2 2020/21

Duration: Two Hours

Candidates must attempt ALL questions.
This is an open book exam but all answers must be in your own words to demonstrate your understanding. You must also show your working.
Answers that do not demonstrate understanding or show working may not receive marks.
Formulae and Tables for Examinations 2002 may be used.
You may use a permitted calculator.

1.

(a) A life insurer is planning to launch a new five year endowment assurance. There is some debate over the final product design so the following results have been produced for two options, A and B.

Option	Profit signature
A	(-1,125, 345, 505, 531, 300)
B	(-603, 146, 256, 224, 451)

The insurer wishes to proceed with the product with the shortest discounted payback period, using a risk discount rate of 9% per annum. Determine which design the insurer should choose.

(5 marks)

Solution: [*Mentioned in class and similar tutorial.*]

Year	A Profit sig (A)	9% discounted	cumulative
1	−1125	−1032.110092	−1032.110092
2	345	290.3795977	−741.7304941
3	505	389.9526574	−351.7778366
4	531	376.1737871	24.39595044
5	300	194.9794159	219.3753663
		219.3753663	

Year	B Profit sig (B)	9% discounted	cumulative
1	−603	−553.2110092	−553.2110092
2	146	122.885279	−430.3257302
3	256	197.6789709	−232.6467593
4	224	158.6872473	−73.95951198
5	451	293.1190552	219.1595432
		219.1595432	

[4]

The insurer would choose contract A because it has a DPP of 4 years compared to contract B which has DPP of 5 years. [1]

(b) Explain why the discounted payback period is a useful profit criterion for the insurer.

(1 mark)

Solution: [*Similar examples and arguments were done in class*]

Insurers only have limited amounts of capital. The DPP measures how quickly that capital is returned to the insurer so that they can invest in selling new policies.

(c) The design that you have identified in (a) will be sold to policyholders aged 35, for a premium of £2,000 payable annually in advance. Calculate the profit margin that the insurer will receive on the contract, on the following basis:

Mortality	AM92 select
Risk discount rate	9% p.a.

(5 marks)

Solution: [*Mentioned in class and similar tutorial.*]

Year	A Profit sig (A)	9% discounted	Premium	q_x
1	−1125	−1032.110092	2000	0.000577
2	345	290.3795977	2000	0.000683
3	505	389.9526574	2000	0.000765
4	531	376.1737871	2000	0.000813
5	300	194.9794159	2000	0.00087
		219.3753663		

Year	prob	discount	prem
1	1	1	2000
2	0.999423	0.91743119	1833.80367
3	0.99874	0.84167999	1681.239616
4	0.997976	0.77218348	1541.241714
5	0.997165	0.70842521	1412.833655
			8469.118655

Therefore,
Profit margin= 2.59%

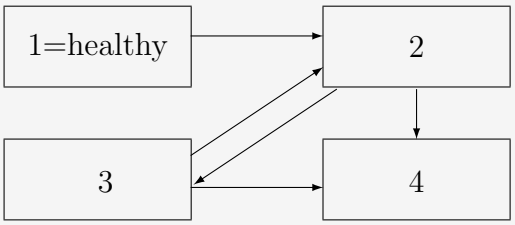
[Total 11 marks]

2. An insurer sells Long-Term Care (LTC) policies. It uses a 4-state Markov model to price these policies in which state 1 is the healthy state and state 4 is the dead state.

   (a) Draw a 4-state Markov model that could be used by an insurer for these policies with appropriate labels for each state and necessary transitional intensity arrows resulting ${}_tp_x^{11} = {}_tp_x^{\overline{11}}$, but ${}_tp_x^{33} \neq {}_tp_x^{\overline{33}}$.

(2 marks)

Solution: [*Lecture Notes and Tutorial.*]



${}_tp_x^{44} = {}_tp_x^{\overline{44}}$, ${}_tp_x^{44} = 1$.

(b) For $j = 2$ and $j = 4$, state with brief reasons whether ${}_tp_x^{jj} = {}_tp_x^{\overline{jj}}$ or ${}_tp_x^{jj} \neq {}_tp_x^{\overline{jj}}$.
(2 marks)

Solution: [*Lecture Notes*]
${}_tp_x^{22} \neq {}_tp_x^{\overline{22}}$ since return is possible and ${}_tp_x^{44} = 1 = {}_tp_x^{\overline{44}}$ since it is not possible to leave the dead state once in the dead state.

(c) State an expression for ${}_tp_x^{31}$ and, without doing any calculations, briefly explain why your expression holds.
(1 mark)

Solution: [*1 mark for sensible comment.*]
it's zero because return to state 1 is impossible.

(d) Suggest, with brief reasons, in what circumstances it might be reasonable for ${}_tp_x^{11} = {}_tp_x^{\overline{11}}$
(1 mark)

Solution: [*1 mark for sensible comment.*]
If there really is no significant chance of recovering from state 2, say if it depends on conditions that more-or-less permanent. Or if the terms of the policy trigger a permanent annuity once state 2 is reached.

[Total 6 marks]

3. A woman aged 59 and her 60 year old husband wish to buy a policy which provides the following benefits:

   - a lifetime annuity starting at £30,000 per annum payable annually in advance while they are both alive, reducing to £20,000 once one of them dies; and
   - an annuity of £2,500 per annum payable annually in advance to each of their twin granddaughters, currently aged 11, commencing on the death of either grandparent and ceasing when the granddaughters reach age 21.

   (a) Assuming the following basis:

Mortality	Man - PMA92C20
	Wife - PFA92C20
	Granddaughters - Nil
Interest	4% per annum
Expenses	1% of the premium

   i. Calculate the granddaughters' benefit as the value of an annuity certain minus a joint-life temporary annuity, or otherwise.

(6 marks)

Solution: [*Similar question was done in class and tutorial. Also, Students can solve with a 9-year or 10-year annuity immediate*]

granddaughters: $5,000\big(\ddot{a}_{\overline{10}|} - \ddot{a}_{60:59:\overline{10}|}\big)$

where

$$\ddot{a}_{\overline{10}|} = \frac{1 - v^{10}}{d} = 8.43533 \qquad [\ 1\ ]$$

and

$$\ddot{a}_{60:59:\overline{10}|} = \ddot{a}_{60:59} - \big(v^{10}\big)\big({}_{10}p_{60:59}\big)\big(\ddot{a}_{70:69}\big) \qquad [\ 1\ ]$$

$${}_{10}p^{m}_{60} = \frac{9238.134}{9826.131} = 0.94016 \qquad [\ 1\ ]$$

$${}_{10}p^{f}_{59} = \frac{9473.745}{9866.200} = 0.96023 \qquad [\ 1\ ]$$

Assuming independance:

$$\ddot{a}_{60:59:\overline{10}|} = 14.248 - \big(v^{10}\big)\big(0.94016\big)\big(0.96023\big)\big(9.932\big) = 8.19073 \qquad [\ 1\ ]$$

Therefore,

$$\text{granddaughters' benefit} = 5,000\big(8.43533 - 8.19073\big) = 1223 \qquad [\ 1\ ]$$

   ii. Calculate the couple's benefit, then use the granddaughters' benefit from part (i) to calculate the single premium to be paid for the policy.

(8 marks)

Solution: [*Similar question was done in class and tutorial*]

EPV of benefits = Single premium (SP) $\times$ (100% − 1%)
= Single premium (SP) $\times$ 99% [1]

Couple: $20,000\big(\ddot{a}_{\overline{60:59}}\big) + 10,000\big(\ddot{a}_{60:59}\big)$ [1]

[Or, $30,000\big(\ddot{a}_{60:59}\big) + 20,000\big(\ddot{a}_{60|59} + \ddot{a}_{59|60}\big)$]

Couple: $20,000\big(\ddot{a}^{\mathrm{m}}_{60} + \ddot{a}^{\mathrm{f}}_{59} - \ddot{a}_{60:59}\big) + 10,000\big(\ddot{a}_{60:59}\big)$ [1]
$= 20,000\big(15.632 + 16.982 - 14.248\big) + 10,000\big(14.248\big)$
$= 20,000\big(18.366\big) + 10,000\big(14.248\big)$
$= 367,320 + 142,480$
$= 509,800$ [2]

Therefore,
Total benefit $= 1223 + 509800 = 511023$ [1]
Total premium $= \dfrac{511023}{0.99} = £516,185$ [1]

(b) Comment on the appropriateness of the decision to ignore the granddaughters' mortality.

(2 marks)

Solution: [*2 marks for sensible comment*]
The mortality of the granddaughters is likely to be very low . Ignoring this mortality slightly overestimates the cost of providing that benefit. Hence, acts as a margin in the pricing basis. That component is $< 0.25\%$ of the total value of benefits so the impact overall premium is very small.

[Total 16 marks]

4. Your department is developing a new unit-linked savings policy, which will not include any guaranteed death benefits.

(a) A marketing colleague has suggested using only a fund management charge to recoup expenses, to make the policy easy to explain. Discuss this suggestion.

(3 marks)

Solution: [*1 mark for each sensible point, additional comments that could receive a mark: challenging to set an appropriate FMC, method hides initial costs from policyholders. Maximum 3 marks.*]

- Would be simple to explain to policyholders and also simple to administer
- Insurer aims for the charges to match its expenses in terms of amount and timing.

- FMC collected will be larger later in the policy term yet expenses will be much greater at the beginning, therefore significant mismatch in timing.
- FMC is collected annually and depends on investment performance and size of the policyholder's funds – will not match fixed expenses.
- If only change FMC it will need to be quite high to cover all the expenses and include margin for prudence due to the mismatched nature – may be difficult to sell with a very high FMC, even if no other charges.

(b) Suggest an alternative structure of charges that may be suitable for this contract, and give a reason for the inclusion of each charge suggested.
(4 marks)

Solution: [*1 mark for each sensible suggestion -max 4. No mark for Mortality charge as there is no guaranteed death benefit.*]
Allocation rate – Percentage of the policyholder premium that will actually be converted to units. May be low in early years then increase, to even more than 100% later in the policy, to offset the insurer's initial expenses.
Bid/offer spread – The policyholder cannot buy and sell units at the same price – buy at the offer price and sell at the bid price – effectively a proportion of each premium is levied as a charge.
Policy fee – a monetary amount dedicated from the policyholder's fund on a regular basis, by selling units from the fund – to match the insurer's regular fixed costs per policy.
Fund Management charge – a regular percentage deduction from the policyholder's fund to meet the investment management expenses of the insurer.
Surrender Penalty – a defined percentage of the unit fund depending on the duration at surrender (only in the early years of the policy to cover initial expenses not recouped elsewhere).

(c) Briefly explain how a unitised with-profits policy differs from a unit-linked savings policy. **(2 marks)**

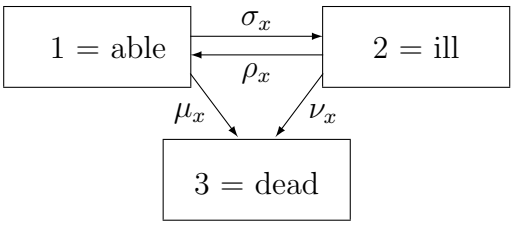
Solution: [*1 mark for each relevant point, , additional comments that could receive a mark: only UWP has bonuses, U/L has potential for higher returns with riskier investments. Maximum 2 marks*]

- Unitised with Profits policy looks similar to a Unit-linked policy in terms of premiums being invested, known charges to be deducted etc.
- However, U/L policies will be invested directly in the asset class selected by the policyholder, and the unit price will fluctuate in line with the actual investment performance, while UWP investments will be determined by the insurer, and the rate of return declared will not necessarily reflect the underlying investment performance – there will be some smoothing of the declared rates.

- UWP may guarantee a minimum investment return (eg 0% pa) but U/L policies do not.

[Total 9 marks]

5. A life insurer uses the following model for pricing its income protection policies:



(a) Write down the equation of value for the following contract sold to a healthy life age x, defining all terms used:

- Benefits of £25,000 per annum are payable continuously while the policyholder is unable to work due to sickness, reducing to £15,000 per annum after one year of benefit payments.
- A maximum benefit payment term of three years applies for any one period of sickness.
- A deferred period of three months applies for all new claims.
- Premiums at rate $\bar{P}$ per annum are payable continously up to age 65 years while healthy, and while sick with duration less than the deferred period.
- Premiums are waived while benefits are payable.
- Interest of $i\%$ per annum effective.
- Expenses are 3% of all premiums payable.

(6 marks)

Solution: [*Mentioned in class and similar tutorial.*]

Equation of value

EPV[Premiums]=EPV[Benefits + Expenses]

$$0.97\overline{P}\int_0^{65-x}\left({}_tp_x{}^{11} + {}_{3|12}\,{}_tp_x{}^{12}\right)\mathrm{dt} \qquad [\ 2\ ]$$

$$= 25000\int_0^{65-x} v^t\left({}_{15|12}\,{}_tp_x{}^{12} - {}_{3|12}\,{}_tp_x{}^{12}\right)\mathrm{dt} \qquad [\ 1.5\ ]$$

$$+ 15000\int_0^{65-x} v^t\left({}_{39|12}\,{}_tp_x{}^{12} - {}_{15|12}\,{}_tp_x{}^{12}\right)\mathrm{dt} \qquad [\ 1.5\ ]$$

where

$${}_{d}\,{}_tp_x{}^{ij} = \mathrm{P}\Big[\text{life is in state } j \text{ at age } x+t, \text{ for duration} \le d \,|\, \text{life is in state } i \text{ at age } x\Big] \qquad [\ 1\ ]$$

(b) The above contract also includes a one month off-period. Explain what this means and why it is included in the policy design.

(2 marks)

Solution: [*2 marks for sensible comment*]

The off-period is the minimum period of good health that must pass before two episodes of sickness are considered as different included in the policy design to prevent policy holders "recovering" briefly to avoid a drop in benefit payments.

[Total 8 marks]

6. Amanda has been a member of her employer's defined benefit pension scheme for 15 years, since she was aged 40. The scheme provides a retirement pension of 1/60th of Final Pensionable Salary for each year of service, where Final Pensionable Salary is the average salary earned over the three years before retirement.

   (a) Given Amanda's salaries have been £32,000, £36,000 and £41,000 in the last three years respectively and assuming there is no allowance for future salary increases, calculate the annual pension Amanda will receive if she stops working now and collects her pension at Normal Retirement Age.

   (2 marks)

   Solution: [*Mentioned in class and similar tutorial.*]

   $$\text{Annual pension } = \frac{15}{60} \times \text{FPS}$$

   where

   $$\text{FPS } = \frac{1}{3}\big(32,000 + 36,000 + 41,000\big) = 36,333$$

   $$\text{AP } = £9083.33$$ [2]

   (b) The scheme guarantees to pay pension benefits for at least five years. On the basis given below, calculate the cost to the pension scheme of providing this guarantee for Amanda, assuming pensions are paid annually in advance and the Normal Retirement Age (NRA) is 65.

Mortality	PFA92C20
Interest	4% per annum

   (3 marks)

   Solution: [*Similar examples and arguments were done in class*]

Cost of benefit without guarantee

$9083.33\ddot{a}_{65} = 9083.33 \times 14.871 = 135,078$ [0.5]

Cost of benefit with guarantee

$9083.33\left(\ddot{a}_{\overline{5}|} + v^5 {}_5p_{65}\ddot{a}_{70}\right)$ [2]

where

$$\ddot{a}_{\overline{5}|} = \frac{1 - v^5}{d} = 4.629895$$

$${}_5p_{65} = \frac{9392.621}{9703.708}$$

$$\ddot{a}_{70} = 12.934$$

Hence,

Cost of benefit with guarantee $= 135,522$

Therefore,

Cost of guarantee $= 135,522 - 135,078 = £444$ [0.5]

[Total 5 marks]

END OF PAPER (TOTAL 55 marks)